

Sri Lanka Institute of Information Technology



Lab Submission
Lab sheet No 05

Exercise 2

IT25100826

Udumullage S.H.U.

IT2011 - Artificial Intelligence and Machine Learning

B.Sc. (Hons) in Information Technology

1. Where does your favourite LLM rank currently?

GPT-5.6 Sol (max)
GPT-5.6 Sol is currently in the 5th place in the leaderboard

Model ↑↓	Features ↔		Intelligence ↔		Price ↔	Speed ↔	Latency ↔	End-to-End Response Time ↔	Further Analysis	
	Context Window ↑↓	Creator ↑↓	Artificial Analysis Intelligence Index ↓		Cost per Task USD ↑↓	Median Tokens/s ↑↓	Latency First Chunk (s) ↑↓	Total Response (s) ↑↓		
Claude Opus 5 (max)	1M	Anthropic	63		\$2.34	56	51.78	60.70	Model ↗	Providers ↗
Claude Opus 5 (xhigh)	1M	Anthropic	63		\$1.80	54	25.15	34.44	Model ↗	Providers ↗
Claude Fable 5 (with fallback)	1M	Anthropic	62		\$3.14	65	95.42	103.15	Model ↗	Providers ↗
Claude Opus 5 (high)	1M	Anthropic	61		\$1.23	55	20.40	29.55	Model ↗	Providers ↗
GPT-5.6 Sol (max)	1M	OpenAI	61		\$0.95	74	140.04	146.81	Model ↗	Providers ↗

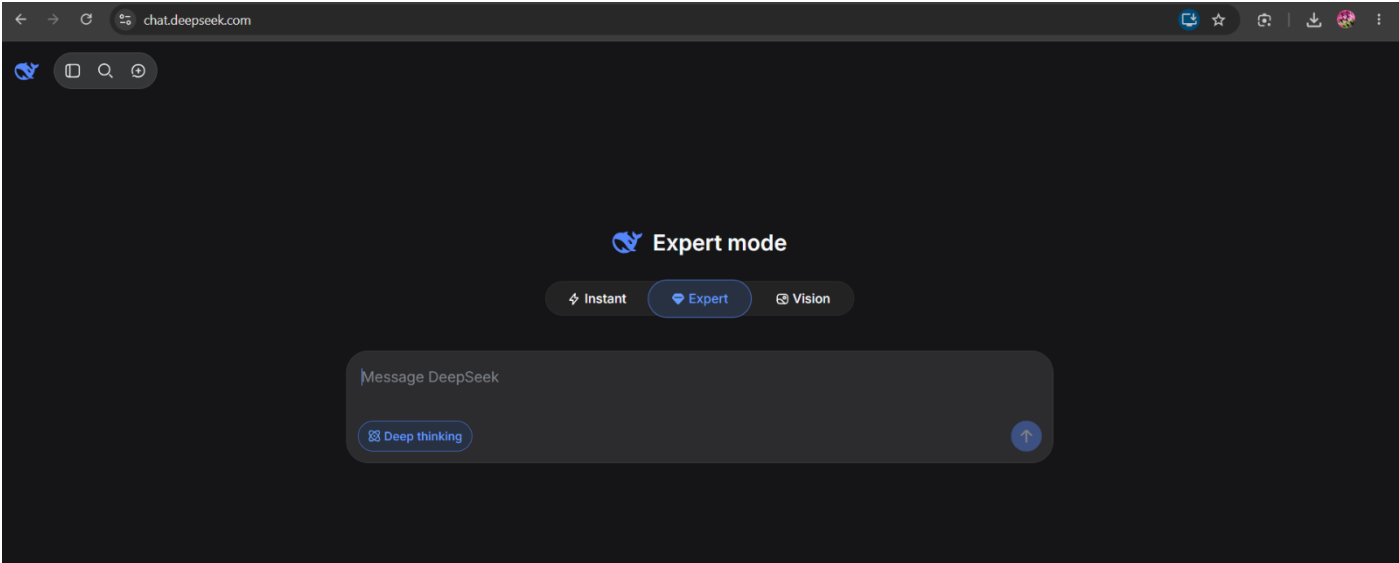
2. Try out at least two models that you haven't tried before

3. Run two prompts of your choice

Prompt 1:
Explain the difference between a database and a DBMS with three examples.

Prompt 2:
Explain machine learning to a beginner using a simple real-life example.

(i) Deepseek



Prompt 1

← → ↺

chat.deepseek.com/a/chat/s/4b8e38e4-4b68-4452-89d...

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Database vs DBMS

💎 Expert

⊕ ↻

Explain the difference between a database and a DBMS with three examples.

📄 ✎

🔗 Thought for 8 seconds ▾

- We need answer. Need explain difference database and DBMS with three examples. We can structure. Database is structured collection of data; DBMS is software managing databases. Examples perhaps: Database examples: a company's customer table, library catalog, hospital patient records. DBMS examples: MySQL, PostgreSQL, Oracle, Microsoft SQL Server, MongoDB, SQLite. Need three examples maybe of each? User says "with three examples" ambiguous: three examples illustrating difference? Could give three pairs or three examples each. Need explain. We can say database is the data itself, DBMS is software. Use analogy: database is like a warehouse contents, DBMS is warehouse management system. Three examples: relational DBs in MySQL vs actual "employees" database; PostgreSQL vs "ecommerce" database; MongoDB vs "product catalog" database. Need ensure answer clear.

Let's craft concise. We can include examples:

1. MySQL is a DBMS; the set of tables for an online store's orders/customers is a database.
2. Oracle Database is a DBMS; a hospital's patient records schema/data is a database.
3. MongoDB is a DBMS; a social media app's user profiles and posts collection is a database.

Maybe mention database = data + schema, DBMS = engine providing CRUD, indexing, transactions, security, concurrency. Need avoid confusion Oracle "Database" product name but it's DBMS. Fine.

Need final with maybe table. Desired oververbosity 5. Use headings.

A database is an organized collection of data, while a DBMS (Database Management System) is the software that manages, stores, retrieves, and secures that data.

Concept	What it is	Analogy
Database	The actual data and its structure	The contents of a warehouse
DBMS	The software that controls access and operations	The warehouse management system

Three Examples

1. **MySQL is a DBMS.**

A database would be the set of tables holding an online store's **customers, orders, and products**.

2. **PostgreSQL** is a DBMS.

A database would be the structured collection of a hospital's **patient records, appointments, and billing data**.

3. **MongoDB** is a DBMS.

A database would be the collection of documents for a social media app's **user profiles, posts, and comments**.

In short:

- **Database** = the data itself
- **DBMS** = the tool used to create, manage, and interact with the database



Message DeepSeek

🔍 Deep thinking



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Prompt 2

Database vs DBMS

Expert

+

🔗

Explain machine learning to a beginner using a simple real-life example.

📄 ✎

🔍 Thought for 4 seconds ▾

- We need to explain machine learning to a beginner with a simple real-life example. The user wants an explanation suitable for a beginner, so avoid jargon. Use analogy. Probably something like teaching a child to recognize animals, or email spam filter, or recommending movies. Let's pick email spam filter as it's common and intuitive. Explain that instead of writing rules manually, we show examples and the system learns patterns. Use simple terms. Structure: introduce ML as a way for computers to learn from examples rather than being explicitly programmed. Then example: spam filter. Show how traditional programming would require writing many rules, but ML learns from labeled emails. Emphasize that the computer finds patterns on its own. Keep it friendly, concise.

Imagine you want to teach a computer to recognize spam emails. Instead of writing a long list of rules like "if the email contains the word 'free' or 'winner', mark as spam," you take a different approach.

You collect thousands of emails that are already labeled as **spam** or **not spam**. You show these examples to the computer and say, "Look at these. Figure out what makes spam different from normal mail."

The computer examines the examples—words, patterns, sender addresses, formatting—and **learns the patterns on its own**. After seeing enough examples, it can look at a brand-new email it has never seen before and predict whether it is spam or not.

That's machine learning in a nutshell:

Instead of programming exact rules, you give the computer data and let it learn the rules by itself.

A simple real-life analogy:

Think of how a child learns to identify a dog. You don't give them a checklist of "has four legs, barks, has fur." You point to different dogs and say "dog," and eventually the child recognizes dogs—even ones they've never seen before—because they learned the pattern from examples. Machine learning works the same way.



Message DeepSeek

 Deep thinking



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Prompt 1

← → ↺ chat.qwen.ai/c/guest ☆ 🖨️ ⬇️ 🌐 ⋮

🗨️ Qwen3.7-Plus ▾

Log in Sign up

Explain the difference between a database and a DBMS with three examples.

📄 ✎

💡 Thinking completed >

To understand the difference between a database and a Database Management System (DBMS), it helps to use a simple analogy: **Think of a library.**

- The **database** is the collection of books (the actual information/data).
- The **DBMS** is the librarian and the library's cataloging system (the software that organizes, retrieves, and manages the books).

Here is the technical breakdown of the difference, followed by three examples.

The Core Differences

Feature	Database	DBMS (Database Management System) 📄
What is it?	A structured collection of data or records.	A software application used to create and manage the database.
Primary Role	To store, hold, and organize the actual data.	To provide an interface for users to access, manipulate, and secure the data.
Nature	Passive (it just sits there holding data).	Active (it processes queries, enforces rules, and handles transactions).
Dependency	Cannot function or be easily accessed without a DBMS.	Can be used to create and manage multiple different databases.

Three Examples to Illustrate the Difference

Example 1: The E-Commerce Platform

- **The Database:** The actual collection of tables containing customer names, email addresses, shipping addresses, and order histories. This is the raw, organized data.
- **The DBMS:** MySQL (or PostgreSQL). This is the software installed on the server. When a

customer clicks "View Order History," the e-commerce website doesn't talk directly to the database; it sends a request to the **MySQL DBMS**, which then fetches the specific data from the database and returns it.

Example 2: The Social Media App

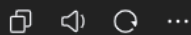
- **The Database:** A massive collection of JSON-like documents containing user profiles, uploaded photos, text posts, and "likes."
- **The DBMS: MongoDB.** Because social media data is highly unstructured and changes frequently, a traditional relational database isn't ideal. **MongoDB** is the NoSQL DBMS software that knows how to store these flexible documents, index them for fast searching, and ensure that when you post a photo, it is saved correctly without crashing the app.

Example 3: The Banking System

- **The Database:** The ledger of financial transactions, account balances, loan records, and customer PINs (stored securely).
- **The DBMS: Oracle Database.** Banks require extreme security, zero data loss, and the ability to handle thousands of transactions per second. **Oracle** is the enterprise-level DBMS software that manages the database. It enforces strict rules (e.g., ensuring that if money is deducted from Account A, it is simultaneously added to Account B) and locks records to prevent two people from accessing the same account data at the exact same millisecond.

Summary

In short, the **database** is the *data itself*, while the **DBMS** is the *software engine* that allows you to create, read, update, delete, and secure that data. You cannot practically use a modern database without a DBMS.



+ Ask Qwen

Thinking ▾



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Prompt 2

Qwen3.7-Plus ▾

Log in

Sign up

Explain machine learning to a beginner using a simple real-life example.

💡 Thinking completed >

Imagine you are hiring a new assistant to help you sort fruit in a grocery store, but this assistant has never seen an apple or an orange before.

You have two ways to teach them. This perfectly explains the difference between traditional programming and Machine Learning (ML).

The Traditional Way (Writing Rules)

You try to write a strict instruction manual for your assistant:

- "If it is red and round, it is an apple."
- "If it is orange and bumpy, it is an orange."

The problem? What if you hand them a *green* apple? Or a smooth, shiny orange? Your strict rules break down, and the assistant gets confused. To fix it, you'd have to constantly write new, more complicated rules for every single exception.

The Machine Learning Way (Learning from Examples)

Instead of giving them a rulebook, you do something different. You put a big basket of mixed apples and oranges in front of them.

1. **You show them examples:** You pick up a fruit, hand it to them, and say, "*This is an apple.*" You do this 50 times. Then you pick up an orange and say, "*This is an orange.*" You do this 50 times.
2. **They find the patterns:** Without you explicitly telling them how, the assistant's brain naturally starts noticing hidden patterns. They realize, "*Hmm, apples are usually smooth and can be red or green. Oranges are usually bumpy and have a distinct smell.*"
3. **They make a prediction:** Now, you hand them a brand-new fruit they have never seen before. Based on the patterns they learned, they confidently say, "*This is an apple!*"

That is Machine Learning. Instead of programming a computer with strict, step-by-step rules, you give it lots of examples (data) and let it figure out the patterns on its own. Once it learns the pattern, it can make accurate guesses about *new* things it has never seen before.

How This Maps to a Computer

- **The Assistant** = The Machine Learning "Model" (the computer program).
- **The Basket of Fruit** = The "Training Data" (the examples you feed the computer).
- **Noticing Smooth vs. Bumpy** = "Feature Extraction" (the computer finding mathematical patterns in the data).
- **Identifying the New Fruit** = "Prediction" or "Inference" (the computer applying what it learned to new data).

Machine Learning in Your Daily Life

You already interact with machine learning every day:

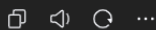
1. **Spam Filters:** Your email doesn't use strict rules to catch spam (because spammers constantly change their wording). Instead, the ML model learned from millions of emails that humans

previously marked as "Spam," noticing patterns like suspicious links or urgent, weird phrasing.

2. **Netflix/Spotify Recommendations:** The system doesn't know your personal taste. But it has learned patterns from your past behavior ("*Users who watched X also watched Y*") and predicts what you might like next.
3. **Smartphone Face ID:** It didn't come pre-programmed with your face. During setup, you showed it your face from different angles (training data). It learned the unique patterns of your features so it can recognize you even if you grow a beard or wear glasses.

The One-Sentence Summary

Machine learning is the science of teaching computers to learn from examples and find patterns, rather than programming them with strict, step-by-step rules.



+ Ask Qwen

Thinking ▾



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Overall Observation

Overall, both DeepSeek and Qwen3.7-Plus provided correct and useful answers to the two prompts. DeepSeek mainly focused on simplicity and explaining concepts through easy real-life examples. Qwen3.7-Plus provided a deeper explanation by combining real-life examples with technical concepts. Therefore, DeepSeek is more suitable for a beginner who wants a quick and simple understanding, while Qwen3.7-Plus is more suitable for a student who wants both a simple explanation and deeper technical knowledge.

Comparison of the Two LLMs

Aspect	DeepSeek	Qwen3.7-Plus
Database & DBMS explanation	Explained the concept using a simple library analogy , making it easy for a beginner to understand.	Used a library analogy too, but gave a more detailed technical breakdown of database vs DBMS.
Database examples	Gave 3 practical examples: e-commerce, social media, and banking.	Also gave 3 practical examples and explained how the DBMS works in each situation.
Technical details	Focused mainly on the basic concept and differences.	Included more technical details such as queries, transactions, indexing, security, and data management .
Machine Learning explanation	Used a fruit-sorting example to explain how ML learns from examples and identifies patterns.	Used the same type of real-life fruit example , then connected it to technical concepts such as training data, feature extraction, model, and prediction .
Real-life applications	Mentioned examples such as spam filters, Netflix/Spotify recommendations, and Face ID.	Also explained several real-life applications and showed how ML is used in everyday systems.
Beginner friendliness	Very easy to understand because the explanation is simple and conversational.	Beginner-friendly, but slightly more technical and detailed.
Level of detail	Moderate and focused on the main idea.	Higher level of detail with additional technical explanations.
Overall style	Simple, direct, and easy to follow.	Detailed, structured, and informative.

